



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Scikit-Learn Basics

DSA0404-Fundamentals of data science

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Introduction



What is Scikit-Learn?

- Open-source Machine Learning library for Python.
- Built on NumPy, SciPy, and Matplotlib.
- Provides simple and efficient tools for:
 - Classification
 - Regression
 - Clustering
 - Model selection
 - Preprocessing



Importing Scikit-Learn



- `from sklearn import datasets`
- `from sklearn.model_selection import train_test_split`
- `from sklearn.linear_model import LinearRegression`

Common Modules

- `sklearn.datasets` → Sample datasets
- `sklearn.model_selection` → Train-test split
- `sklearn.preprocessing` → Scaling and encoding
- `sklearn.metrics` → Accuracy, MSE, etc.



Machine Learning Workflow



- Load dataset
- Split data into training and testing sets
- Choose model
- Train model
- Predict output
- Evaluate performance

Workflow Diagram

➤ Dataset → Preprocessing → Train/Test Split → Model
Training → Prediction → Evaluation



Example – Linear Regression



```
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
X, y = make_regression(n_samples=100, n_features=1, noise=10)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = LinearRegression()
model.fit(X_train, y_train)
pred = model.predict(X_test)
print(pred[:5])
```

Purpose

- Predict a continuous value such as sales, marks, or price.

Model Evaluation

```
from sklearn.metrics import mean_squared_error,  
r2_score  
  
mse = mean_squared_error(y_test, pred)  
  
r2 = r2_score(y_test, pred)  
  
print("MSE:", mse)  
  
print("R2 Score:", r2)
```

Important Metrics

- **MSE (Mean Squared Error):** Lower is better.
- **R² Score:** Closer to 1 is better.



Data Preprocessing in Scikit-Learn



```
from sklearn.preprocessing import StandardScaler  
  
scaler = StandardScaler()  
  
X_scaled = scaler.fit_transform(X)
```

Why Preprocessing?

- Removes scale differences between features.
- Improves model performance.
- Helps algorithms converge faster.



Applications of Scikit-Learn

- Student performance prediction
- House price prediction
- Spam email detection
- Customer segmentation
- Medical diagnosis support
- Fraud detection

Advantages and Limitations

Advantages

- Easy API
- Extensive documentation
- Fast for small and medium datasets
- Many built-in algorithms

Limitations

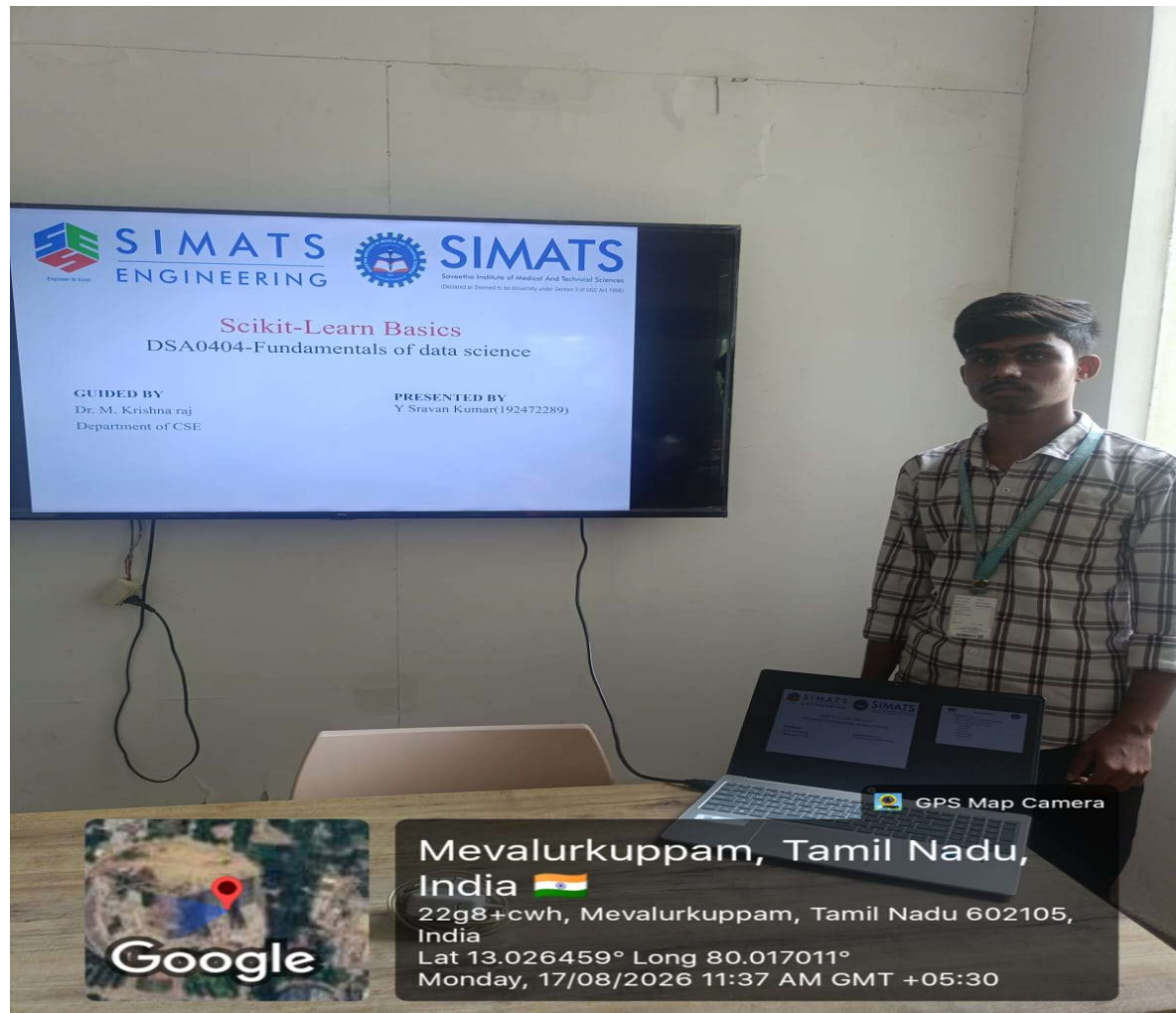
- Not ideal for deep learning
- Less efficient for very large datasets
compared with specialized frameworks



Conclusion



- Scikit-Learn is a beginner-friendly machine learning library.
- Supports preprocessing, training, prediction, and evaluation.
- Suitable for academic projects and industry prototypes.



THANK YOU!